1. Compare and contrast the File System and Database Management System (DBMS) in terms of data organization, redundancy, consistency, security, data sharing, and data independence with suitable examples.
2. Explain the architecture of the DBMS and data model concept.
3. Explain the characteristics/ features of DBMS.
4. Construct an Entity-Relationship (ER) diagram for a Hotel Management system, library management system,/ hospital management system, /University Database Management System ,/
5. Classify and explain the different types of attributes used in the Entity-Relationship (ER) model with suitable examples.
6. Explain the concepts of Generalization, Specialization, and Aggregation in the Enhanced Entity-Relationship (EER) model with suitable diagrams and examples.
7. Describe the three-schema architecture of a Database Management System (DBMS). Explain the concepts of data abstraction and logical as well as physical data independence with neat diagrams.
8. Discuss the roles and responsibilities of the Database Administrator (DBA). Also explain the different categories of database users in a database environment.
9. Explain the significance of different keys with suitable example.
10. explain the procedure for converting a multivalued attribute into a relational schema with suitable example.
11. Explain the set operations in Relational Algebra. Illustrate the operations of Union, Intersection, Difference, and Cartesian Product using suitable examples.
12. Explain the Relational Data Model and discuss the concepts of relation, tuple, attribute, domain, degree, cardinality, and relational schema with suitable examples.
13. Explain the process of mapping an Entity-Relationship (ER/EER) model to a Relational Model. Illustrate the mapping rules for entities, relationships, and multivalued attributes with suitable examples.
14. Differentiate between Unary and Binary Relational Algebra operations. Demonstrate their application by writing suitable relational algebra expressions for a relational database.